

Image caption

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Abstract

Your abstract.

1 Introduction

Image caption generation aims to generate a natural language description of the given image. While covering both Computer Vision and Natural Language Processing, it can be regarded as a cross-modal task.

In 1.1, I describe the pipeline to process images and texts from dataset. In 1.2, I introduce the framework of Encoder-Decoder Model. IN 1.3, I describe the scheduled sampling method I used to mitigate the exposure bias that occurs between the training and inference phases.

1.1 Pipeline

The pipeline of image captioning is shown in Figure 1

The Encoder-Decoder framework for image captioning involves using a ResNet model to encode images into feature maps, which are then used by an LSTM decoder to generate captions. Attention mechanisms are employed to enhance the model's focus on relevant image features.

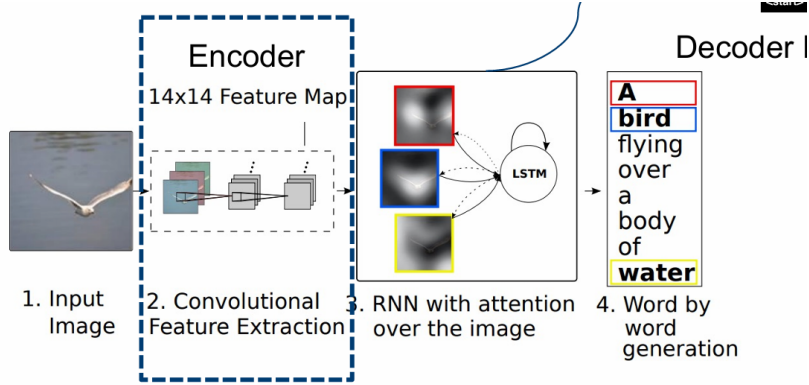


Figure 1: Pipeline

1.2 Model framework

1. The encoder uses ResNet101 pretrained on the ImageNet dataset to extract a 14×14 feature map with 2048 channels from the input image.
2. The visual attention mechanism integrates the image feature vector g from encoder with the previous hidden state h_{t-1} from decoder to compute attention weights. The weight $\alpha_{t,i}$ for each time step t and spatial location i is calculated as follows:

$$\alpha_{t,i} = \frac{\exp(MLP(g_i, h_{t-1}))}{\sum_k \exp(MLP(g_k, h_{t-1}))} \quad (1)$$

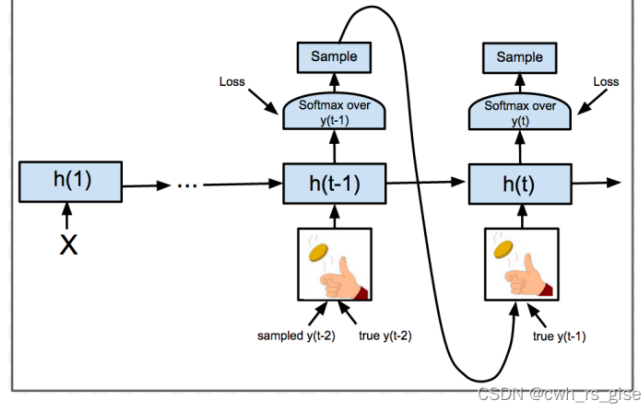


Figure 2: Schedule sampling

The attention weighted feature map z_t is then derived by aggregating the product of the attention weights and the corresponding image features:

$$z_t = \sum_{i=1}^L \alpha_{t,i} \cdot g_i \quad (2)$$

3. The decoder also consider the textual information by weighting the the last hidden state by β . The weight β_t for each time step t is calculated as follows:

$$\beta_t = \frac{\exp(\text{Linear}(h_{t-1}))}{\sum_k \exp(\text{Linear}(h_{t-1}))} \quad (3)$$

4. Then the decoder as a LSTM architecture produces the next word in the caption and the hidden state by combining the current word embedding with the attention weighted image features and attention weighted last hidden state as follows:

$$\begin{aligned} c_t &= \beta_t \cdot z_t \\ &= \beta_t \sum_{i=1}^L \alpha_{t,i} \cdot g_i \end{aligned} \quad (4)$$

1.3 scheduled sampling

Scheduled sampling was introduced to address the issue of "exposure bias" which occurs in RNNs trained entirely with the correct outputs (teacher forcing) struggle during inference when they must rely on their own predictions.

During training, scheduled sampling alternates between using the correct previous output and the model's own prediction from the previous step by a designed probability p , which typically decreases during the training process. The structure of schedule sampling is shown in Figure 2

Typically, the schedule follows one of these patterns:

- **Linear decay:**

$$p(t) = \max(0, 1 - kt)$$

- **Exponential decay:**

$$p(t) = e^{-kt}$$

- **Inverse sigmoid decay:**

$$p(t) = \frac{1}{1 + e^{k(t-t_0)}}$$

Here, k is a rate of decay, and t_0 indicates the midpoint in the sigmoid decay schedule. Figure 3 shows the decay patterns above.

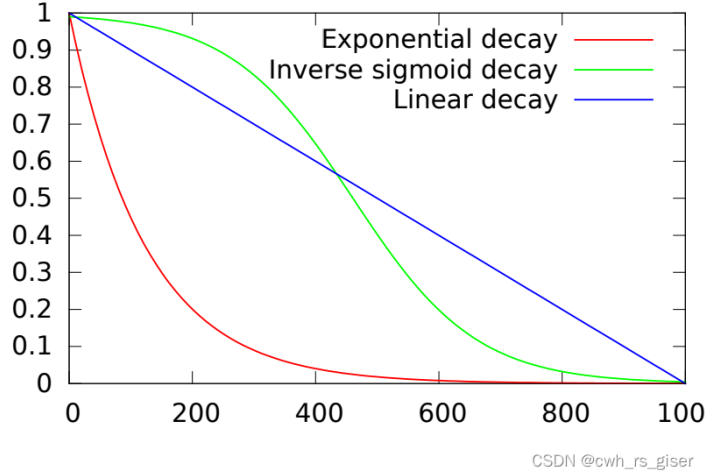


Figure 3: Schedule sampling patterns

2 Experiment

I use Flickr8k as dataset. For pre-processing, I resize all images to 256×256 , and randomly apply horizontal flip with probability of 0.5, and normalize them such that their means are (0.485, 0.456, 0.406) and standard variances are (0.229, 0.224, 0.225) for each channel. With batch size=128 and zero padding, Resnet101 encode each image to the feature map with size of (14, 14, 2048). The embedding dimension is 300, attention dimension is 256 and decoder size is 256. I employ RMSprop as optimizer with initial learning rate as 0.001. All experiments are executed within 20 epochs.

2.1 Evaluation

To thoroughly assess the performance of our model, I utilize several key metrics, each catering to different aspects of language translation and generation. These metrics are:

- **BLEU (Bilingual Evaluation Understudy)**: This metric measures how many words and phrases in the machine-generated translations match reference translations, focusing on precision.
- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation)**: ROUGE evaluates how much of the content in reference texts is captured by the generated outputs, emphasizing recall.
- **METEOR (Metric for Evaluation of Translation with Explicit ORdering)**: METEOR enhances evaluation by considering synonymy, morphology, and sentence structure.
- **CIDEr (Consensus-based Image Description Evaluation)**: Developed for image captioning, CIDEr measures the similarity of generated descriptions to multiple reference descriptions.
- **SPICE (Semantic Propositional Image Caption Evaluation)**: SPICE assesses the semantic fidelity of captions by comparing the propositional content with that of reference captions.
- **SPIDEr**: Combines SPICE and CIDEr to provide a holistic measure of both semantic relevance and consensus accuracy.

In this practice, I compute the scores for each evaluation method and sample both the best and the worst prediction to manually analysis model’s performance.

2.2 Result

I implemented the linear decay with $k=(5, 10, 20)$, exponential decay with $k = (0.97, 0.99)$, inverse sigmoid decay with $k=(5, 10)$ to optimize the basic model.

The results are shown in Table 1.

model	Bleu-1	Bleu-2	Bleu-3	Bleu-4	Rouge	METEOR	CIDEr	SPICE	SPIDEr
baseline	0.607	0.419	0.277	0.175	0.413	0.197	0.487	0.140	0.313
linear 5	0.569	0.370	0.232	0.145	0.394	0.180	0.418	0.126	0.272
linear 10	0.585	0.385	0.245	0.154	0.408	0.186	0.445	0.134	0.289
linear 20	0.627	0.414	0.265	0.167	0.414	0.187	0.458	0.137	0.297
exp 0.95	0.628	0.423	0.272	0.171	0.419	0.184	0.475	0.132	0.303
exp 0.97	0.614	0.416	0.273	0.175	0.415	0.190	0.476	0.138	0.307
sigmoid 5	0.620	0.415	0.267	0.167	0.415	0.188	0.474	0.135	0.304
sigmoid 10	0.609	0.416	0.276	0.177	0.421	0.194	0.492	0.138	0.315
cycle 5	0.616	0.422	0.278	0.181	0.414	0.197	0.490	0.144	0.317
cycle 10	0.632	0.418	0.269	0.173	0.413	0.183	0.460	0.133	0.297
cycle 20	0.609	0.397	0.252	0.156	0.421	0.180	0.429	0.127	0.278

Table 1: Schedule sampling optimization

- **Bleu Scores:** The ‘cycle 10’ configuration scores highest in Bleu-1, indicating it captures the most relevant terms effectively. Conversely, ‘cycle 5’ excels in Bleu-3 and Bleu-4, suggesting better performance in longer phrase matching.
- **Rouge:** The ‘sigmoid 10’ configuration stands out with the highest Rouge score, implying superior content matching from the references.
- **METEOR and CIDEr:** The best performances in METEOR and CIDEr are observed in ‘cycle 5’, reflecting its overall translation quality and relevance.
- **SPIDEr:** ‘cycle 5’ also leads in SPIDEr, which combines SPICE and CIDEr, indicating a balanced assessment in terms of both semantic and syntactic elements.

These results highlight the varied effectiveness of different sampling strategies, with cyclic variations showing notable benefits in certain conditions.

2.3 Cycle decoy

Cycle decay schedule is based on sinusoidal function. The details of this schedule is given by:

$$p = \frac{1}{2} \left(1 + \cos \left(\frac{2\pi x}{k} \right) \right) \quad (5)$$

The oscillation of p enables the model to continuously adapt and improve through iterative learning and self-correction.

The higher k is, the faster the oscillation moves.

I tested $k=(5, 10, 10)$ for cycle decoy. The results, detailed in Table 1, indicate that $k = 5$ yields the best performance.

2.4 Subjective observation

I sample both the best and the worst predictions of each evaluation metrics subjectively evaluate the performance of the best cycle 5 model. The details are shown in the Figure 4

- **Bleu Score:** This metric emphasizes exact word matches between the predicted and reference captions. The high score for the dog with the blue ball (0.99) suggests a strong lexical alignment, while the near-zero score for the bear image mislabeled as a dog highlights its sensitivity to exact word mismatches.
- **Rough Score:** This metric accounts for the order of words and their occurrence. It rewards sequences that match more comprehensively. The perfect score (1.0) for the white dog in the snow reflects an exact match, illustrating its effectiveness for straightforward scenes. In contrast, the zero score for the mislabeled bears underscores its incapacity to handle incorrect semantic content.

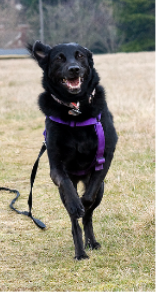
	good	prediction	bad	prediction
Bleu		'dog is playing with a blue ball .', 0.99		'dog is shaking off water .', 2.02e-16
Rough		'white dog is running through the snow 1.0		dog is shaking off water 0.0
METEOR		white dog is running through the snow 0.56		baby with a baby in a bib and a baby 0.02
CIDEr		black dog with a purple collar is running through the grass 4.48		dog is shaking off water 0.0

Figure 4: Sample

- **METEOR Score:** METEOR improves upon Bleu by considering synonyms and stemming, offering a balance between precision and recall. The moderate score (0.56) for the dog in the snow indicates reasonable but imperfect caption accuracy, suggesting that METEOR is moderately forgiving of phrasing variations but still demands semantic correctness.
- **CIDEr Score:** Focused on rewarding unique and relevant terms, CIDEr scored high (4.48) for the accurate depiction of the black dog. This indicates its strength in recognizing and rewarding detailed, descriptive captions, particularly in distinguishing relevant from common words.

3 Conclusion

I presented an extensive analysis of various decay strategies for scheduled sampling in the training of an image captioning model. The experimental results indicate that different decay schedules, such as linear, exponential, and cycle decay, significantly affect the model’s performance in terms of language generation and alignment with image content.

Specifically, the cycle decay schedule emerged as particularly effective, showing superior performance across multiple metrics, including BLEU, ROUGE, METEOR, and SPIDEr.

This suggests that incorporating periodic adjustments in the training process can help the model better generalize from training data to real-world scenarios, thereby mitigating exposure bias more effectively.